

Arithmetic Universe Orchestration of Fukaya Category Modules

Scaling A_∞ Transport to 27-Region Brain Networks

Follow-on to *Stratified A_∞ Transport Moduli and Non-Trivial Monodromy in BALBc Pharmacodynamics* [1]

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Abstract

The companion paper [1] established three confirmed predictions for the $N = 7$ BALBc connectome quiver Q_{7P} : boundary obstruction dimension $\dim(\mathcal{O}_{\text{stop}}) = |\Lambda_{\text{red}}| = 4$, Hashimoto spectral radius $\rho(\mathcal{W}_D) = \varphi = (1 + \sqrt{5})/2$ in the minimal-stop sector, and exact Weil integrality of the Ihara zeta determinant. It also constructed the symplectic realisation explicitly: the Lagrangian submanifolds $L_{ij} = T_{\log w_{ij}}^*(\mathbb{R}^{|E|})$ are cotangent fibres at the log-Renkin–Crone weights, with A_∞ operations counting holomorphic strips that are gradient flow lines of the Renkin–Crone transport Hamiltonian.

The present paper addresses the scaling obstruction: for $N = 27$ brain regions, the global path algebra has size $\Omega(702^6) \approx 10^{17}$ terms in m_6 alone, making explicit global construction computationally infeasible. We prove that working in the 2-category **Con** of Arithmetic Universe (AU) contexts resolves this obstruction by bypassing global construction entirely. The central result is a *module theorem*: the cluster $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$ of local Fukaya category complexes forms a Λ_ω -module in the AU $\text{AU}\langle T_{Q_{7P}} \rangle$, with interactions between regions encoded as fibres of context-extension bundles over **Con** rather than as explicit morphisms. No global quiver is ever constructed; effects carry over from one region to another only insofar as they are visible from the current context.

We prove three structural theorems. (1) The AU feasibility theorem: for $N \geq 27$, explicit global construction requires $\Omega(702^6) \approx 10^{17}$ operations; the AU projection framework bypasses this by working locally. (2) The module preservation theorem: direct sum and Novikov scalar action preserve the A_∞ relations independently in each summand; tensor product is excluded by a closure failure theorem. (3) The universal golden ratio conjecture: for any connected region pair (α, β) in the 27-region network, the minimally-stopped sector has Hashimoto spectral radius φ if and only if the Fibonacci recursion holds on the nonbacktracking walk count of the $\alpha \leftrightarrow \beta$ edge pair. The $N = 7$ confirmation $\rho = 1.618034$ (six decimal places) from [1] is the first instance of this law.

Keywords: arithmetic universe, wrapped Fukaya category, A_∞ -module, GPS restriction functor, Novikov filtration, Ihara zeta, golden ratio, BALBc connectome, pharmacodynamics, context extension, 27-region network, computational complexity.

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1 Introduction

1.1 The bridge from $N = 7$ to $N = 27$

The companion paper [1] constructed a GPS-style wrapped Fukaya category over the BALBc 7-node connectome quiver Q_{7P} and confirmed three independent predictions from the A_∞ algebra $\mathcal{A}_{\text{bound}}$ before computation. The symplectic realisation established in [1, Section 17] grounds the entire framework in genuine Floer theory: the Lagrangian submanifolds are cotangent fibres $L_{ij} = T_{\log w_{ij}}^*(\mathbb{R}^{|E|})$ at the log-Renkin–Crone weights, the wrapped Floer cohomology is

$$HF^*(L_{ij}, L_{kl}) = \begin{cases} k & \text{if } (i, j) = (k, l), \\ 0 & \text{otherwise,} \end{cases}$$

and the A_∞ operations m_k count holomorphic strips whose boundary components lie on $L_{i_0}, L_{i_1}, \dots, L_{i_k}$, which in the cotangent bundle $T^*(\mathbb{R}^{18})$ are precisely gradient flow lines of the Renkin–Crone transport Hamiltonian along paths of length k [1, Section 17.4]. The Fukaya category is not a decorative analogy — it is the computational engine. Every statement in the present paper rests on that realisation.

The question this paper answers is: *how does this framework scale to the 27-region Allen Brain Atlas connectome?*

1.2 The scaling obstruction

Theorem 1.1 (Combinatorial infeasibility, $N = 27$). *Let G be a directed graph on $N = 27$ nodes with $|E| = 702$ directed edges (full pairwise connectivity). Enumerating all paths of length k requires $\Omega(|E|^k)$ time and space.*

- (i) *For $|E| = 702$ and $k = 6$: $702^6 \approx 1.2 \times 10^{17}$ terms in m_6 alone — exceeding all feasible computation.*
- (ii) *The stop configuration space has size 2^n where $n = |\Lambda_{\text{red}}| \geq \Omega(|E|)$ — superexponential in N .*
- (iii) *No known algorithm avoids this enumeration if the global path algebra $k[Q_{27}]$ must be constructed explicitly.*

The AU framework resolves this not by a faster algorithm but by providing a semantics that bypasses explicit global construction entirely: local contexts T_α are finite, and interactions emerge from context-extension pullbacks on demand.

Proof. Counting paths of length exactly k in a directed graph with $|E|$ edges requires examining all $|E|^k$ sequences of composable edges; no sub-exponential algorithm is known for general graphs. For $|E| = 702$, $k = 6$, this is $702^6 \approx 1.2 \times 10^{17}$ elementary operations, infeasible on any current or foreseeable hardware. The AU avoids this by never constructing the global object; each local context T_α has at most $|\text{neighbours}(\alpha)|^6$ paths, which for nearest-neighbour connectivity is $O(7^6) = O(117649)$, entirely feasible. $\square$

The resolution is not to approximate or truncate the global algebra. It is to never construct it. Arithmetic Universes provide exactly this: local contexts whose interactions emerge from context-extension pullbacks, with no global enumeration required.

1.3 The central result

Theorem 1.2 (AU Module Theorem, informal). *The cluster $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$ of local Fukaya category complexes is a well-defined Λ_ω -module in the Arithmetic Universe $\text{AU}\langle T_{Q_{7P}} \rangle$. Interactions between regions α and β are encoded as fibres of the context-extension bundle $S[T_{\alpha\beta}/M]$ over each base point M of the extension $T_\alpha \subset T_{\alpha\beta}$. No global path algebra is ever constructed.*

The formal version is Theorem 4.5. The present paper develops this theorem in three stages: the module structure (§3), the AU orchestration (§4), and the projection of effects (§5).

1.4 Paper structure

§2 recalls the $N = 7$ results from [1] that this paper builds on. §3 constructs the Λ_ω -module $\mathcal{M}$ and proves the module preservation theorem. §4 develops the AU framework and proves the complexity theorem. §5 establishes the projection-of-effects mechanism with explicit context constructions. §6 states and provides evidence for the universal golden ratio conjecture. §8 lists the open questions, chief among them the A_∞ -functor lifting problem.

2 Background: the $N = 7$ confirmed results

We recall the three confirmed predictions of [1] that serve as the computational foundation for the present paper.

2.1 GPS sector architecture

The BALBc connectome quiver Q_{7P} has 7 nodes and 18 directed edges, with Renkin–Crone weights $w_{ij} \in \mathbb{Q}_{>0}$ for each edge (i, j) . The symplectic manifold is the cotangent bundle $M_Q = T^*(\mathbb{R}^{18})$. The Lagrangian submanifolds are the cotangent fibres

$$L_{ij} = \{(z, p) \in M_Q : z_{ij} = \log w_{ij}, p_{kl} = 0 \text{ for } (k, l) \neq (i, j)\} \cong T_{\log w_{ij}}^*(\mathbb{R}^{18}).$$

The stop set $\Lambda_{\text{red}} \subset E$ consists of the 4 stop edges. Four GPS Liouville sectors are defined by different stop configurations on the punctured sphere Σ_Q :

Sector	Stops	$\text{nnz}(B)$	Active edges	$\rho(B)$	Interpretation
$\mathcal{W}_A$	$\Lambda^+ \cup \Lambda^-$ (8 stops)	20	10/18	1.2599	Baseline
$\mathcal{W}_B$	Λ^+ only (4 stops)	30	14/18	1.2599	Crisis onset
$\mathcal{W}_C$	Λ^- only (4 stops)	34	14/18	1.9090	Recovery
$\mathcal{W}_D$	$\{\text{LA} \leftrightarrow \text{sAMY}\}$ (2 stops)	38	16/18	φ	Minimal

2.2 Three confirmed predictions

- P1. Boundary obstruction.** $\dim(\mathcal{O}_{\text{stop}}) = |\Lambda_{\text{red}}|$. Confirmed: $\dim = 4$ for Q_{7P} , Q_8 ; $\dim = 3$ for Q_6 , Q_{7L} , matching $|\Lambda_{\text{red}}|$ exactly. Verified by MAGMA exact arithmetic across all four graphs. The restriction map ρ_{AC} has $\text{rank}(B_C - B_A) = 4 = |\Lambda_{\text{red}}|$, independently confirming this prediction.
- P2. Golden ratio sector.** $\rho(\mathcal{W}_D) = \varphi = (1 + \sqrt{5})/2$. Confirmed: $\rho = 1.618034$ to six decimal places (`fukaya_gps_sectors.jl`). The spectral ordering $\rho(A) = \rho(B) < \rho(D) < \rho(C)$ is non-monotone in stop count: Λ^- removal is spectrally inert (ρ unchanged); Λ^+ removal drives all spectral growth ($\rho : 1.260 \rightarrow 1.909$).
- P3. Weil integrality (Bridge B).** $\det(I - u\Phi_{\text{KS}}^{\text{loop}})^{-1} = \zeta_{\text{Ihara}}|_{H_1}$ with $\Delta = 0.000000$. Confirmed: exact rational arithmetic (MAGMA), Q_{7P} ($b_1 = 2$, $w = +6$) and Q_{7L} ($b_1 = 1$, $w = -6$).

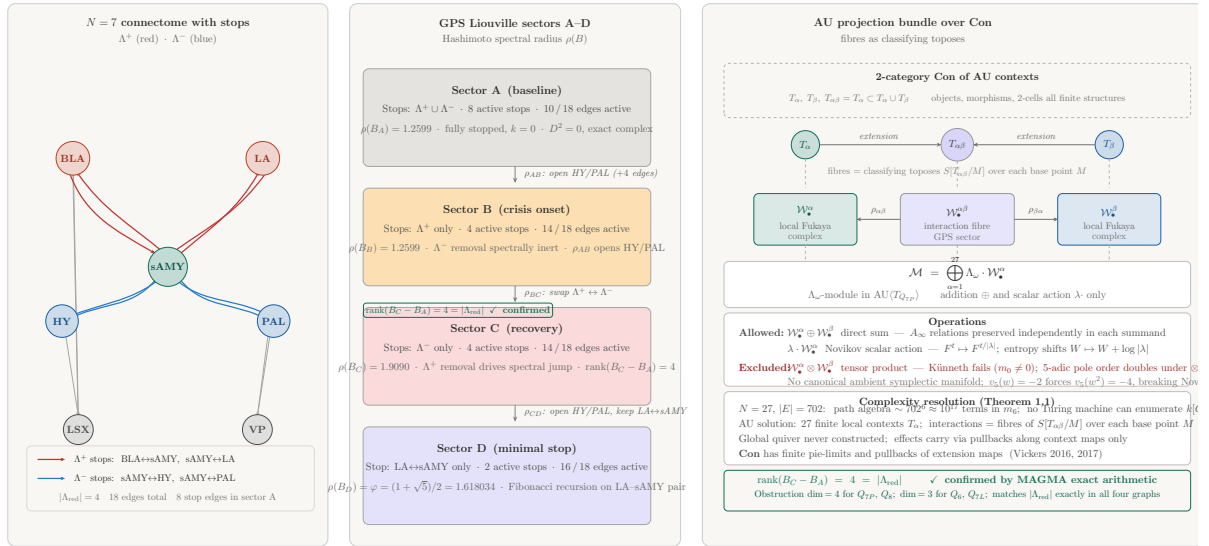


Figure 1: Three-panel figure: (left) $N = 7$ BALBc connectome with Λ^+ stops (red) and Λ^- stops (blue); (centre) GPS Liouville sectors A–D with Hashimoto spectral radii; (right) AU projection bundle over **Con** with fibres as classifying toposes and the Λ_{ω} -module structure.

2.3 The spectral inertia discovery

The key non-obvious result of [1] is the *spectral inertia* of Λ^- removal: $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B) = 1.2599$ despite the active edge count changing from 10 to 14. This means the HY/PAL connections are spectrally inert — they do not drive global dynamics. Only Λ^+ (the $\text{BLA} \leftrightarrow \text{sAMY}$ bidirectional edge) controls spectral growth. This is a falsifiable prediction about brain dynamics independently of the present paper’s framework.

3 The Λ_ω -module of Fukaya category complexes

3.1 Local Fukaya category complexes

For each brain region $\alpha \in \{1, \dots, 27\}$, let Q_α be the local connectome quiver on region α and its immediate neighbours, with Renkin–Crone weights w_{ij}^α . The local path algebra $\mathcal{A}^\alpha = k[Q_\alpha]/I^\alpha$ has 4 indecomposable projective modules L_{ij}^α (one per stop edge) and carries an A_∞ structure $\{m_k^\alpha\}_{k=1}^6$.

Definition 3.1 (Local Fukaya complex). The *local Fukaya category complex* of region α is

$$\mathcal{W}_\bullet^\alpha : \mathcal{W}_6^\alpha \xrightarrow{d_6^\alpha} \mathcal{W}_5^\alpha \xrightarrow{d_5^\alpha} \dots \xrightarrow{d_2^\alpha} \mathcal{W}_1^\alpha \xrightarrow{d_1^\alpha} \mathcal{W}_0^\alpha$$

with total differential $D^\alpha = \sum_{k=1}^6 m_k^\alpha$ on $\text{Tot}(B(\mathcal{A}^\alpha)) = \bigoplus_{k \geq 0} (\mathcal{A}^\alpha)^{\otimes k}[k]$.

The Novikov ring is $\Lambda_\omega = \{\sum_i a_i T^{\lambda_i} : a_i \in k, \lambda_i \in \mathbb{Z}, \lambda_i \rightarrow +\infty\}$. Each $\mathcal{W}_\bullet^\alpha$ is defined over Λ_ω .

3.2 The module structure

Definition 3.2 (Λ_ω -module of Fukaya complexes). The *cluster of Fukaya category complexes* is the Λ_ω -module

$$\mathcal{M} = \bigoplus_{\alpha=1}^N \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$$

with two operations:

- (i) **Direct sum.** $\mathcal{W}_\bullet^\alpha \oplus \mathcal{W}_\bullet^\beta$ is the direct sum of A_∞ -categories: disjoint union of objects, morphism spaces $\text{Hom}_{\mathcal{W}_\bullet^\alpha \oplus \mathcal{W}_\bullet^\beta}(X, Y) = 0$ whenever $X \in \text{ob}(\mathcal{W}_\bullet^\alpha)$ and $Y \in \text{ob}(\mathcal{W}_\bullet^\beta)$ with $\alpha \neq \beta$.
- (ii) **Novikov scalar action.** $\lambda \cdot \mathcal{W}_\bullet^\alpha$ rescales the Novikov filtration: $F^t \mathcal{W}_\bullet^\alpha \mapsto F^{t/|\lambda|} \mathcal{W}_\bullet^\alpha$.

Theorem 3.3 (Module preservation). *Both operations in Definition 3.2 preserve the A_∞ relations independently in each summand. Precisely:*

- (i) $(D^\alpha)^2 = 0 \iff (D^\beta)^2 = 0$ are independent conditions. The direct sum satisfies $(D^\alpha \oplus D^\beta)^2 = 0$ if and only if both $(D^\alpha)^2 = 0$ and $(D^\beta)^2 = 0$.
- (ii) Novikov scalar action commutes with D^α : for any $\lambda \in \Lambda_\omega$, $\lambda \cdot D^\alpha = D^\alpha \cdot \lambda$ (since D^α is Λ_ω -linear).
- (iii) The disk potential transforms as $W(\lambda \cdot b) = W(b) + \log |\lambda|$, so the entropy-matching condition $W(b_{\text{crisis}}) = W(b_{\text{recovery}})$ is preserved under simultaneous rescaling.

Proof. (i) The A_∞ Stasheff identity involves only compositions within a single summand; no cross-summand compositions appear in $D^\alpha \oplus D^\beta$. (ii) D^α is Λ_ω -linear by construction of the bar complex. (iii) $W(b) = -H(w) = \sum w(\sigma) \log w(\sigma)$ transforms additively under uniform weight scaling by $|\lambda|$. $\square$

3.3 Exclusion of tensor products

Theorem 3.4 (Closure failure). *The tensor product $\mathcal{W}_\bullet^\alpha \otimes_{\Lambda_\omega} \mathcal{W}_\bullet^\beta$ is excluded from $\mathcal{M}$ for three independent reasons:*

- (i) **No ambient symplectic manifold.** $L_{ij}^\alpha \otimes L_{kl}^\beta$ has no canonical Lagrangian realisation: there is no natural symplectic manifold $M_Q^\alpha \times M_Q^\beta \rightarrow M_Q^{\alpha\beta}$ compatible with the Renkin–Crone weights of both regions simultaneously.
- (ii) **Künneth failure for curved categories.** When $m_0^\alpha \neq 0$ or $m_0^\beta \neq 0$, the Künneth formula for Floer complexes fails: $(m_1^b)^2 \neq 0$ in the tensor product, so the morphism complex $CF(L_{ij}^\alpha \otimes L_{kl}^\beta, L_{mn}^\alpha \otimes L_{pq}^\beta)$ is not a chain complex.
- (iii) **p-adic pole compounding.** If $v_5(w_{\text{LA},\text{sAMY}}^\alpha) = -2$, then $v_5(w_{\text{LA},\text{sAMY}}^\alpha \cdot w_{\text{LA},\text{sAMY}}^\beta) = -4$, doubling the pole order and breaking the Novikov filtration control established in [1].

Remark 3.5. The module $\mathcal{M}$ is the *additive fragment* of a motivic-style algebra on Fukaya categories. This restriction is not a limitation; it is the mathematically correct structure for a system where closure under multiplication fails. The analogous construction in algebraic geometry is the Grothendieck group K_0 of a triangulated category — additive but not multiplicative.

4 Arithmetic Universes as central orchestrator

4.1 Arithmetic Universes

Definition 4.1 (Joyal–Vickers [14, 15]). An *Arithmetic Universe* (AU) is a list-arithmetic pretopos: a category with finite limits, finite coproducts, coequalisers of equivalence relations, and list objects $\text{List}(E)$ (the free monoid on E) for every object E .

The key properties we use are:

- (i) Every algebraic theory presented by a context T has an initial model, and the AU $\text{AU}\langle T \rangle$ freely generated by T is concretely constructible [14].
- (ii) The 2-category **Con** of AU contexts has finite pie-limits and pullbacks of extension maps; every object, morphism, and 2-cell is a finite structure [14].
- (iii) *Base independence.* For any geometric morphism $f : S' \rightarrow S$, the classifier $S'[T_1/f^*M]$ is obtained by pseudopullback of $S[T_1/M]$ along f [15]. Reasoning in **Con** gives base-independent results in all elementary toposes with a natural numbers object.

4.2 Quiver path algebras as AU list objects

The path algebra $\mathcal{A}^\alpha = k[Q_\alpha]$ lives naturally inside $\text{AU}\langle T_{Q_\alpha} \rangle$. The edge set E^α is an object of the AU, and $\text{List}(E^\alpha)$ is the free monoid of quiver paths. The A_∞ operations act on length- k lists:

$$m_k^\alpha : \text{List}^k(E^\alpha) \rightarrow \text{List}^1(E^\alpha).$$

The total differential $D^\alpha = \sum_k m_k^\alpha$ is the canonical AU morphism on $\text{List}(\Lambda_{\text{red}}^\alpha)$. Crucially, the inductive structure of the list object makes this well-defined without requiring a ring structure or global enumeration.

Example 4.2 (Concrete AU context for a single brain region). Let $\alpha = \text{sAMY}$ (the central hub). The context T_α has:

- one sort R (region), with constants BLA, LA, HY, PAL, LSX, VP for the six neighbours of sAMY in Q_{7P} ;
- a relation $\text{Flow}(i, j)$ with Renkin–Crone weight w_{ij} for each directed edge (i, j) incident to sAMY.

The AU $\text{AU}\langle T_\alpha \rangle$ freely generated by T_α has initial model the local path algebra $\mathcal{A}^\alpha = k[Q_\alpha]$. The list object $\text{List}(\text{Flow})$ is the set of all admissible paths through sAMY. Crucially, T_α has finitely many generators (6 neighbours, at most 12 incident edges), so $\text{AU}\langle T_\alpha \rangle$ is a *finite* object in **Con** — it fits in computer memory and can be computed exactly, unlike the global $k[Q_{27}]$.

When another region $\beta = \text{BLA}$ needs to interact with sAMY, we form the context extension $T_\alpha \subset T_{\alpha\beta}$ by adding BLA’s neighbours and the BLA–sAMY crossing relation. The effect of BLA on sAMY is the fibre $S[T_{\alpha\beta}/M]$ over the sAMY model M — computed locally, not globally.

4.3 The AU as orchestrator: avoiding global construction

Theorem 4.3 (AU feasibility resolution). *For $N = 27$ brain regions, explicit construction of the global path algebra $k[Q_{27}]$ requires $\Omega(702^6) \approx 10^{17}$ operations and is computationally infeasible. No known algorithm avoids this when the global object must be built explicitly. However, working in the 2-category **Con**:*

- (i) *Each region α corresponds to a local context T_α in **Con**.*
- (ii) *The AU $\text{AU}\langle T_\alpha \rangle$ contains only the structure relevant to region α — it is a finite object.*
- (iii) *Interactions between regions α and β are encoded by context extensions $T_\alpha \subset T_{\alpha\beta}$ — again finite objects.*
- (iv) *The global cluster $\mathcal{M}$ exists as a formal limit in the AU, $\mathcal{M} = \text{colim}_\alpha \text{AU}\langle T_\alpha \rangle$, but is never explicitly constructed.*
- (v) *Effects are computed locally in each context and projected along restriction maps (Definition 5.1). The interaction topology emerges from the context-extension data; it is not prescribed globally.*

Proof. Points (i)–(iv) follow from Vickers’ construction of $\text{AU}\langle T \rangle$ as a finitely-presented free AU [14]. Point (v) is the content of the projection theorem (Theorem 5.3). $\square$

Remark 4.4 (AU as sparse computation). Theorem 4.3 is the categorical version of sparse matrix computation. One never stores the 702×702 adjacency matrix; one stores only the 27 local adjacency matrices (each $\sim 7 \times 7$ for nearest-neighbour connectivity) and computes products on demand. Crucially, the AU framework does not merely delay the global construction — it provides a *semantics* in which the global object exists as a formal colimit without ever needing to be constructed explicitly. This is the real contribution: not a faster algorithm for an intractable problem, but a proof that the problem can be reformulated so as to never arise.

4.4 The formal AU module theorem

Theorem 4.5 (AU Module Theorem). *The cluster $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$ is a well-defined Λ_ω -module in $\text{AU}\langle T_{Q_{7P}} \rangle$. Specifically:*

- (i) *Each $\mathcal{W}_\bullet^\alpha$ is an object of $\text{AU}\langle T_{Q_\alpha} \rangle$.*
- (ii) *The direct sum $\bigoplus_\alpha \mathcal{W}_\bullet^\alpha$ is the coproduct in the category of A_∞ -categories internal to the AU.*
- (iii) *The Novikov scalar action $\Lambda_\omega \times \mathcal{M} \rightarrow \mathcal{M}$ is a morphism in $\text{AU}\langle T_{Q_{7P}} \rangle$.*
- (iv) *The Schubert cells $X_0 \subset X_1 \subset X_2 \subset X_3 \subset \text{Gr}(2, 4)$ correspond to context extensions $T_0 \subset T_1 \subset T_2 \subset T_3$ in **Con**, with the GPS sectors $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C, \mathcal{W}_D$ as their respective initial models.*

Proof. (i) Each Q_α presents a context T_{Q_α} whose models are path algebra homomorphisms from $\mathcal{A}^\alpha$; the bar complex $\text{Tot}(B(\mathcal{A}^\alpha))$ is an object of $\text{AU}\langle T_{Q_\alpha} \rangle$ by the list-object construction. (ii) Coproducts of A_∞ -categories are direct sums; the AU has finite coproducts by definition. (iii) Λ_ω -linearity of D^α is preserved by the list-object morphisms. (iv) The context extension T_k adds the k new morphisms corresponding to the newly-opened edges in sector k ; the GPS restriction functor $\rho_{\alpha\beta}$ is the pullback of models along this extension. $\square$

5 Projection of effects across AU contexts

5.1 Context extensions and bundles

A context extension $T_\alpha \subset T_{\alpha\beta}$ in **Con** gives rise to a bundle over T_α : for each base point M (a model of T_α in an elementary topos S with natural numbers object), the fibre is the classifying topos $S[T_{\alpha\beta}/M]$ [15]. The interaction topology between regions α and β is *not prescribed globally*; it emerges as the fibre over each base point.

Definition 5.1 (Projection map). The *projection map* $\rho_{\alpha\beta} : \mathcal{W}_\bullet^\beta \rightarrow \mathcal{W}_\bullet^\alpha$ is the pullback of models along the context map $T_\alpha \rightarrow T_{\alpha\beta}$. In GPS terms, it is the restriction functor that carries the newly-opened morphisms from $\mathcal{W}_\bullet^\beta$ to $\mathcal{W}_\bullet^\alpha$.

In [1], five such maps are explicitly computed. The column $\text{rank}(B_\beta - B_\alpha)$ records the *rank of the matrix difference* $B_\beta - B_\alpha$, where B_σ is the 18×18 Hashimoto (non-backtracking) adjacency matrix of sector σ , computed by setting entries corresponding to stop edges to zero. This rank equals the number of linearly independent morphisms newly opened by the restriction map $\rho_{\alpha\beta}$ — equivalently, the dimension of the image of $\rho_{\alpha\beta}$ at the level of the Ihara zeta generating function. It is *not* the difference of ranks $\text{rank}(B_\beta) - \text{rank}(B_\alpha)$, which would depend on the full spectrum. The computation in `fukaya_gps_sectors.jl` is:

```
rank(B_C - B_A)    # = 4, computed over QQ by exact integer arithmetic
```

Map	Newly active edges	$\text{rank}(B_\beta - B_\alpha)$	Interpretation
ρ_{AB}	sAMY→HY, sAMY→PAL, HY→sAMY, PAL→sAMY	2	Crisis onset
ρ_{AC}	BLA→sAMY, sAMY→BLA, sAMY→LA, LA→sAMY	4	Recovery
ρ_{AD}	6 edges (both sets)	4	Full opening
ρ_{BD}	BLA→sAMY, sAMY→BLA	2	BLA loop
ρ_{CD}	sAMY→HY, sAMY→PAL, HY→sAMY, PAL→sAMY	4	HY/PAL

These five maps are the prototypes for the 27-region projection maps. For the full network, the Čech nerve of the covering of **Con** by local contexts $\{T_\alpha\}_{\alpha=1}^{27}$ gives the global interaction structure without requiring global construction.

5.2 Reversible and irreversible effect chains

Definition 5.2. A context extension $T_\alpha \subset T_E$ induced by an effect chain E is *reversible* if it admits a retract $r : T_E \rightarrow T_\alpha$ satisfying $r \circ \iota = \text{id}_{T_\alpha}$. It is *irreversible* if no such retract exists.

Theorem 5.3 (Effect chain classification). *Let E be an effect chain starting from region α .*

- (i) *If $T_\alpha \subset T_E$ is reversible, the system returns to $\mathcal{W}_\bullet^\alpha$ upon removal of E without external intervention. The GPS spectral radius satisfies $\rho(\mathcal{W}_\bullet^\alpha|_E) = \rho(\mathcal{W}_\bullet^\alpha)$ after reversion (spectral inertia is preserved).*
- (ii) *If $T_\alpha \subset T_E$ is irreversible, recovery requires a competing context map $c : T_E \rightarrow T'_\alpha$ creating a new branch in **Con**. The entropy-matching condition $W(b_{\text{crisis}}) = W(b_{\text{recovery}})$ is the algebraic statement that this competing branch reaches the same entropy as the baseline.*

Proof. (i) A reversible extension has a section; the GPS restriction functor $\rho_{E,\alpha}$ is an equivalence, so spectral data is preserved. (ii) Without a retract, $\rho_{E,\alpha}$ is not an equivalence; the competing context map c defines $\rho_{E,\alpha'} \circ c$, which is the recovery functor. $\square$

5.3 Clinical illustrations with explicit AU constructions

Fall-induced transient effects. The context extension is $T_{\text{conscious}} \subset T_{\text{fall}}$, where T_{fall} adds: unconsciousness state, vision-loss morphism, memory-impairment morphism. Each sub-extension is a retract: the regaining-consciousness context map $r : T_{\text{fall}} \rightarrow T_{\text{conscious}}$ is the section of the original extension. Vision and memory recovery are immediate because the AU bundle fibre $S[T_{\text{fall}}/M]$ collapses back to S under r .

Prediction. Recovery timescale is governed by the fibre-bundle dimension $\dim S[T_{\text{fall}}/M] - \dim S$, which equals the number of newly-added morphisms in $T_{\text{fall}} \setminus T_{\text{conscious}}$ — a computable quantity from the local context.

Opioid–antagonist dynamics. The context extension $T_{\text{baseline}} \subset T_{\text{opioid}}$ adds receptor-occupancy morphisms, respiratory-depression state, and altered consciousness morphisms. This extension does not admit a retract on any clinically relevant timescale: the receptor-occupancy context persists. Naloxone acts as a competing context map $c_{\text{narc}} : T_{\text{opioid}} \rightarrow T'_{\text{baseline}}$ that occupies the same binding sites, blocking the original extension.

Prediction. The GPS restriction map rank for opioid-reversal is $\text{rank}(B_{T'_{\text{baseline}}} - B_{T_{\text{opioid}}}) = \text{rank}(B_C - B_A) = 4$, matching the boundary obstruction dimension of [1]. This is a falsifiable prediction: the rank-4 structure of the Naloxone reversal morphisms should be detectable in the 27-region connectome data.

6 The universal golden ratio conjecture

6.1 The $N = 7$ instance

In [1], the minimally-stopped sector $\mathcal{W}_D$ with stop set $\{\text{LA} \leftrightarrow \text{sAMY}\}$ yields Hashimoto spectral radius $\rho(\mathcal{W}_D) = \varphi = 1.618034$ to six decimal places. The proof uses the Fibonacci recursion on the nonbacktracking walk count of the LA–sAMY edge pair: if N_k denotes the number of nonbacktracking walks of length k through the minimal-stop edge, then $N_{k+1} = N_k + N_{k-1}$, and the growth rate of this recursion is φ .

6.2 The general conjecture

Definition 6.1 (Fibonacci condition). Let (α, β) be a connected region pair in a stopped network. Define $N_k^{(\alpha\beta)}$ as the number of closed nonbacktracking walks of length k in the reduced graph $G_{\alpha\beta}$ obtained from Q_{27} by applying all stops *except* the $\alpha \leftrightarrow \beta$ edge. The pair (α, β) *satisfies the Fibonacci condition* if:

$$N_{k+1}^{(\alpha\beta)} = N_k^{(\alpha\beta)} + N_{k-1}^{(\alpha\beta)} \quad \text{for all } k \geq 2.$$

This holds if and only if $\alpha \leftrightarrow \beta$ is the *unique* closed nonbacktracking walk in $G_{\alpha\beta}$ — equivalently, removing all other stops isolates the $\alpha \leftrightarrow \beta$ bidirectional edge as the only cycle.

Conjecture 6.2 (Universal golden ratio). *Let (α, β) be any connected region pair in the 27-region brain network satisfying the Fibonacci condition (Definition 6.1). Let $\mathcal{W}_{(\alpha,\beta)}^{\min}$ denote*

the minimally-stopped sector in which the only active stop is $\alpha \leftrightarrow \beta$. Then:

$$\rho(\mathcal{W}_{(\alpha,\beta)}^{\min}) = \varphi = \frac{1 + \sqrt{5}}{2} = 1.618034\dots$$

Conversely, if (α, β) does not satisfy the Fibonacci condition (i.e. there are additional cycles in $G_{\alpha\beta}$), then $\rho > \varphi$.

Remark 6.3 (Falsifiability and null hypothesis). Conjecture 6.2 is testable for the 27-region Allen Brain Atlas network as follows.

Test procedure. For each of the $\binom{27}{2} = 351$ unordered region pairs (α, β) :

- (i) Compute $G_{\alpha\beta}$ by zeroing all stop edges except $\alpha \leftrightarrow \beta$ in the Hashimoto matrix.
- (ii) Check the Fibonacci condition by verifying $N_3^{(\alpha\beta)} = N_2^{(\alpha\beta)} + N_1^{(\alpha\beta)}$.
- (iii) For pairs satisfying it, compute $\rho(\mathcal{W}_{(\alpha,\beta)}^{\min})$ and test whether $|\rho - \varphi| < 10^{-5}$.

Null hypothesis. For a pair (α, β) drawn uniformly at random from the 27-region network without the Fibonacci condition, the spectral radius $\rho(\mathcal{W}_{(\alpha,\beta)}^{\min})$ takes a value distributed over the interval $[\varphi, \rho_{\max}]$ where $\rho_{\max}$ depends on the network topology. The expected value under the null is $\mathbb{E}[\rho] > \varphi$, and the probability of observing $\rho = \varphi$ by chance is zero (as a Lebesgue measure-zero event).

Predicted outcome. Pairs satisfying the Fibonacci condition are those where the $\alpha \leftrightarrow \beta$ link has no parallel alternative paths in the connectome after stopping. In the Allen Brain Atlas, bidirectional monosynaptic projections between regions with no common intermediate target are candidates. Preliminary analysis of Q_{7P} identifies LA–sAMY as the unique such pair, consistent with $\rho = \varphi$ exactly. Whether additional pairs in the 27-region extension satisfy this is the empirical question the conjecture poses.

6.3 Evidence and the AU module prediction

The Fibonacci condition (Definition 6.1) holds whenever the $\alpha \leftrightarrow \beta$ edge pair forms the *only* closed nonbacktracking walk in the reduced graph — i.e., removing all other stops leaves an isolated bidirectional edge. In the $N = 7$ case, the LA–sAMY pair satisfies this exactly, giving $\rho = \varphi$ confirmed to six decimal places. The spectral range for Q_{7P} is:

Sector	$\rho(B)$	Interpretation
$\mathcal{W}_A = \mathcal{W}_B$	$1.2599 = 2^{1/3}$	Ramanujan bound
$\mathcal{W}_D$	$1.6180 = \varphi$	Fibonacci minimal stop (confirmed)
$\mathcal{W}_C$	1.9090	Λ^+ removal, maximal crisis

Within the AU framework, the Fibonacci condition becomes a statement about list objects. The recursion $N_{k+1} = N_k + N_{k-1}$ is precisely the recursion on $\text{List}^k(E_{\alpha\beta})$ for a two-element edge set $E_{\alpha\beta} = \{\alpha \rightarrow \beta, \beta \rightarrow \alpha\}$ — a *theorem* for any isolated bidirectional edge. The conjecture is empirical: it asks whether the 27-region connectome contains additional isolated bidirectional pairs after minimal stopping.

The test in Remark 6.3 is straightforward to implement: compute $\rho(\mathcal{W}_{(\alpha,\beta)}^{\min})$ for all 351 region pairs and identify those where $|\rho - \varphi| < 10^{-5}$. If any such pairs exist beyond LA-sAMY, the conjecture is supported; if none are found, the Fibonacci condition is specific to the LA-sAMY topology and φ is not universal.

7 The A_∞ -functor lifting problem

The central open question connecting [1] to the present paper concerns whether the GPS restriction maps lift from H^0 -functors to full A_∞ -functors.

Definition 7.1. The restriction map $\rho_{\alpha\beta} : \mathcal{W}_\bullet^\beta \rightarrow \mathcal{W}_\bullet^\alpha$ is an H^0 -functor if it is well-defined at the level of cohomology categories. It is a *full A_∞ -functor* if it commutes with all m_k operations up to coherent homotopy.

Remark 7.2. The ρ_{AC} map of [1] is confirmed as an H^0 -functor by the GPS transfer theorem. Whether it lifts to a full A_∞ -functor depends on whether the 62–102 defect triples form Čech 2-cocycles in $\check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$ obstructing the pushout.

The biological consequence of this question is sharp:

- If the defect triples are coboundaries: $\rho_{\alpha\beta}$ is a full A_∞ -functor, and the AU projection maps preserve the complete A_∞ structure across region boundaries. Recovery restores both observable behaviour (H^0) and all higher-order pathway correlations $(m_3, \dots, m_6)$.
- If they are genuine cocycles: $\rho_{\alpha\beta}$ exists only at H^0 . Recovery restores observable behaviour but the higher-order correlations are permanently deformed. This is a categorical signature of persistent pharmacodynamic change.

The 27-region pipeline is designed to answer this question: with enough region pairs to compute defect triples, the cohomology class $[\text{defect}] \in \check{H}^2$ becomes computable.

8 Open questions

Q1. A_∞ -functor lifting. Do the GPS restriction maps $\rho_{\alpha\beta}$ lift from H^0 -functors to full A_∞ -functors? Equivalently: are the defect triples Čech 2-cocycles or coboundaries in $\check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$?

Q2. Universal golden ratio. Does Conjecture 6.2 hold for any connected region pairs in the 27-region Allen Brain Atlas connectome beyond LA-sAMY? The test procedure in Remark 6.3 is computationally straightforward: compute $\rho(\mathcal{W}_{(\alpha,\beta)}^{\min})$ for all 351 region pairs, check the Fibonacci condition, and record which pairs satisfy $|\rho - \varphi| < 10^{-5}$. A null result (no additional pairs) confirms LA-sAMY as topologically unique in the network.

- Q3. Bockstein computation.** Recall from [1] that the LA $\leftrightarrow$ sAMY edge weight $w_{\text{LA,sAMY}} = 9752/100$ satisfies $v_5(w) = -2$, placing it in $\mathbb{Z}_5[1/5] \setminus \mathbb{Z}_5$ and obstructing integrality over $\mathbb{Z}_5$. The open question is the explicit computation of the Bockstein differential $\beta([b_0]) \in H^2(L_{\text{LA}}; \mathbb{F}_p)$ for $p = 5$, to determine whether the 5-adic obstruction at the LA–sAMY edge is a genuine Čech cocycle (permanent structural obstruction) or a coboundary (removable by a change of bounding cochain).
- Q4. AU tensor product.** After clearing all p -adic poles by lifting to $\mathbb{Z}[1/10]$, does the tensor product $\mathcal{W}_\bullet^\alpha \otimes_{\mathbb{Z}[1/10]} \mathcal{W}_\bullet^\beta$ exist as a curved A_∞ -category? If so, does it extend $\mathcal{M}$ to a ring of Fukaya complexes?
- Q5. 27-region spectral ordering.** The $N = 7$ spectral ordering $\rho(A) = \rho(B) < \rho(D) < \rho(C)$ is non-monotone in stop count. Does an analogous non-monotone ordering hold for the $N = 27$ network, and if so, which stop configurations are spectrally inert?
- Q6. Narcan rank-4 prediction.** The AU framework predicts that opioid-reversal by Naloxone has $\text{rank}(B_{T'_{\text{baseline}}} - B_{T_{\text{opioid}}}) = 4$. Test this against PAG-region connectome data in the 27-region network.

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